Artifact Abstract

On the Reliability of Code Comprehension Proxies

Paper title

On the Reliability of Code Comprehension Proxies

Link to the accepted paper

https://arxiv.org/abs/2605.23008

Purpose

This artifact contains the data, scripts, web application, and study materials used in the paper. It supports full reproduction of the paper's quantitative results: the correlation analyses between student comprehension proxies (accuracy, timing, self-reported difficulty) and expert-ranked ground truth (Figures 3–11), the task taxonomy from open coding (Figure 1, Table 1), the frequency of comprehension measure types used across prior studies (Table 2), the impact of participant background and fatigue on proxy reliability (Table 7), cross-institution consistency between University 1 and University 2 (Table 8), and the evolution of expert agreement across the three Delphi consensus rounds (Table 4).

Badge

We are applying for the **Artifacts Available**, **Artifacts Functional**, and **Artifacts Reusable** badges.

- **Available**: the artifact is permanently archived on Zenodo with a DOI, independent of the authors' personal infrastructure or the GitHub repository.
- **Functional**: the artifact is documented, consistent, complete, and exercisable, and directly supports the claims made in the paper. Every quantitative claim (Figures 1, 3–11, Tables 1, 2, 3, 4, 6, 7, 8) is mapped in the README to the exact command or script that reproduces it, and `python run_all.py` exercises the full pipeline end to end in under 5 minutes.
- **Reusable**: the artifact is self-contained and reproducible end-to-end via Docker with a single command (`python run_all.py` inside the analysis container), completing in under 5 minutes. It includes all raw data, all analysis scripts, the web application used to run the study, all study materials in both machine-readable and human-readable form, and inter-rater agreement scripts. Full justification is in the STATUS file.

Technology skills assumed by the reviewer, and hardware requirements

Reviewers need only basic familiarity with Docker (`docker load` / `docker run`) to reproduce all quantitative results; no Python, Java, or Maven knowledge is required for that path. Building from source (optional) additionally assumes basic Python and Java/Maven familiarity. No special hardware is required — a standard laptop or desktop suffices. The Docker images are provided as native builds for both x86_64 (amd64) and ARM64 (Apple Silicon) — no emulation required on either platform. Full details are in the REQUIREMENTS file.

Provenance

The artifact is archived on Zenodo: https://zenodo.org/records/19348389 (this version's record). The concept DOI **10.5281/zenodo.19348388** represents all versions and always resolves to the latest one. The artifact is also mirrored (non-archival, convenience only) on GitHub at `github.com/erfan-arvan/on-the-reliability-of-code-comprehension-proxies-replication`.

Instructions

1. Install Docker Desktop (includes Docker Compose v2). No other software required.
2. Download `artifact.zip` from the Zenodo record, extract it, and load the two Docker images matching your machine's architecture (check with `uname -m`; see REQUIREMENTS for details):

   ```
   unzip artifact.zip
   cd artifact
   docker load < code-comprehension-analysis-amd64.tar.gz   # or -arm64
   docker load < code-comprehension-web-amd64.tar.gz         # or -arm64
   ```

3. Smoke test (expected output: `OK`):

   ```
   docker run --rm code-comprehension-analysis:latest python -c "import pandas,
   scipy, matplotlib; print('OK')"
   ```

4. Reproduce all quantitative results (Figure 1, Tables 1, 2, 3, 4, 6, Figures 3–11, Tables 7–8) in under 5 minutes:

   ```
   mkdir -p results plots
   docker run --rm \
     -v "$(pwd)/results:/analysis/results" \
     -v "$(pwd)/plots:/analysis/plots" \
     code-comprehension-analysis:latest python run_all.py
   ```

   Generated CSVs and plots land directly in `results/` and `plots/` on the host; `--rm` means this can be re-run any number of times with no cleanup needed.
5. (Optional) Run the web application used to conduct the study:

   ```
   docker run --rm -p 8080:8080 code-comprehension-web:latest
   ```

   then open `http://localhost:8080` (see `App/README.md`, inside `artifact.zip`, for demo credentials).

Full step-by-step instructions, the dataset schema, and the artifact layout are in `README.MD`, included both standalone in this Zenodo record and inside `artifact.zip`.